INCH-POUND

MIL-T-52984B(ME)
12 September 1988
SUPERSEDING
MIL-T-52984A(ME)
30 August 1984

MILITARY SPECIFICATION

TRUCK, FIRE FIGHTING: BRUSH, 1200 GALLON WATER TANK, 6 X 6, COMMERCIAL CHASSIS

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel-engine-driven (DED), three axle, all-wheel-drive, fire fighting truck having pump and roll capability to fight brush/grass fires, and the ability to provide supporting operations for structural fires.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel Oil, Diesel.
GGG-S-62	- Saw, Chain, Portable, Power.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-3859	- Grease Gun, Hand-Operated, Lever, Push and Screw Type.
MIL-E-10062	- Engine, Preparation for Shipment and Storage of.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables For Inspection By Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-281	- Automobile, Trucks, Truck Tractors, and Trailer Dollies, Preservation and Packaging of.
MIL-STD-454	- Standard General Requirement for Electronic Equipment.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MIL-STD-1410	- Methods of Selection of Industrial Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS24207	- Coupling, Grease Gun; Hydraulic Type Nozzle.

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HANDBOOKS

MILITARY

MIL-HDBK-113 - Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Vehicles Safety Standards and Regulations.
Motor Carrier Safety Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Code of Federal Regulations, Title 45, Subtitle A, part 1201 Motor Vehicle Air Pollution Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management Control District, 9150 Flair Drive, El Monte, CA 91731.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI/ASME	- Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1	- Welding Code Structural - Steel.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 380	- Cleaning and Descaling Stainless Steel Parts, Equipment and Systems.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

Std. No. 114	- Standard for Break Mandrel Blind Rivets.

(Application for copies should be addressed to the Industrial Fasteners Institute, 1717 East Ninth Street, Cleveland, OH 44114.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 414	- Aircraft Rescue and Fire Fighting Vehicles.
NFPA 1901	- Standard for Automotive Fire Apparatus.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION (TRA)

Tire and Rim Association Yearbook, with Military Supplement.

(Application for copies should be addressed to the Tire and Rim Association, 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck shall be a three-axle, all-wheel-drive, DED truck cab-and-chassis. The truck shall be fitted with a fire truck body, a 250 gallon per minute (gpm) minimum fire pump driven by an auxiliary diesel engine; a 1200 usable (minimum) gallon (gal) removable water tank, hosebeds, booster hose reel; operator's instrument panel for fire pump; and such other equipment and accessories as specified herein.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 4.1.2).

3.3.1 Material deterioration prevention and control. The truck shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the truck may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the truck may be newly fabricated from recovered materials to the maximum extent practicable, provided the truck produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the truck.

3.4 Design.

3.4.1 Truck chassis. The truck chassis shall be a three-axle, all-wheel-drive, front-wheel-steer, dual-rear-wheel, conventional type, with three-man cab.

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3.4.2 Weight, ratings, and dimensions.

3.4.2.1 Weight rating. The total gross weight of the complete truck shall not exceed the manufacturer's gross vehicle weight (gvw) rating when fully equipped as specified herein, including all specified fire fighting equipment, serviced with fuel, water and lubricants; loaded with not less than 1200 gal of water in the water tank; not less than a 50 pound (lb) simulated load in each equipment cabinet; rate of hose installed and a crew of three in the cab weighing not less than 175 lbs each or equivalent deadweight. Weight distribution ratio between the front and rear axle shall be such as to ensure positive traction at all wheels and positive steering at the front wheels under all operating conditions specified herein. Weight distribution shall not exceed the manufacturer's rating of individual components.

3.4.2.2 Dimensions. The truck overall dimensions including mounted equipment shall not be more than the following:

a. Width - 8 feet (ft)
b. Length - 26 ft
c. Height - 9 ft

3.4.3 Frame. Chassis frame shall be the manufacturer's standard welded, riveted, or bolted construction, and shall be rated for operation at rated gvw. When operating the fully equipped and loaded truck under conditions specified herein, the frame shall show no evidence of permanent deformation or distortion. Prior to production of the first truck, the contractor shall obtain certification from the original truck chassis manufacturer that:

a. The chassis frame is suitable for the intended operational use and for the mounting of all required fire fighting components specified herein.
b. Any frame modification, including machining, shall be approved and certified by the truck chassis manufacturer.
c. Flame cutting of the frame for modification shall not be allowed.

Certification shall be made available for review by the contracting officer or designated representative.

3.4.3.1 Angles of approach and departure. The angles of approach and departure shall be not less than 40 degrees.

3.5 Construction.

3.5.1 Standard truck, components, and accessories. Except as specified herein, the truck, components, and accessories shall be standard or optional items that meet or exceed the requirements of this specification. All items shall be as represented and rated in the contractor's sales information, including special or mounted equipment. Sales information shall be limited to specifications and technical material, identical to that furnished to the authorized contractor representative for selection of truck models and components. Sales information and special drawings necessary to identify the

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required truck and components shall be on file in the engineering offices of the contracting activity prior to delivery of the items. Truck, components, assemblies, and accessories to be provided hereunder are standard or optional commercial products for the purpose of this specification. Such items shall be construed as commercial motor vehicle products, complying with all DOT Federal Motor Vehicle Safety Standards and Regulations applicable on the date of manufacture.

3.5.2 Interchangeability. All truck parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.5.3 Maintainability.

3.5.3.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, shall be adequate to perform the intended function of assisting or conducting maintenance operations.

3.5.3.2 Ease of maintenance. All major assemblies, including pumps, transmission, hose reels, water tank, and similar items, shall be arranged for individual removal and replacement without major disassembly or removal of other components of the truck body. Covers or plates which must be removed for component adjustment or maintenance shall be equipped with quick-disconnect fasteners. All access covers shall be labeled with nomenclature to indicate what is accessed through them in accordance with MIL-STD-1472, 5.9.9.3 as specified in 3.7. Design for maintainability shall comply with 3.7. Drain outlets shall be located for accessibility.

3.6 Performance characteristics. The fully equipped truck and equipment loaded as specified shall be capable of performing the following:

a. Parking brakes shall hold the truck on a 30-percent grade, in both ascending and descending positions, with the fire pump operating and discharging 80 percent of the water tank capacity at a flow rate of not less than 100 gpm at a pressure of not less than 100 psi through the 1-1/2 inch hose.
b. Service brakes shall hold the truck on a 50-percent grade in both ascending and descending positions.
c. Service brakes shall bring the truck to a stop within a braking distance of 30 ft from a speed of 20 mph on a dry, smooth, level surface free of loose material.
d. Maintaining a maximum speed of not less than 55 mph on dry, level, paved roadway.
e. Accelerating on a level road from a standing start to a speed of 35 mph in 25 seconds.

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f. Maintaining a sustained speed when ascending a 50-percent grade in low-speed range.
g. There shall be no evidence of body distortion, water leakage, tire and body contact, malfunction of components, irregular chassis noise, vibration or sway when subjected to road test of 200 miles over paved roadway at speeds up to 55 mph; 50 miles over graded dirt roadway at not less than 25 mph; and 25 miles off highway over cross-country terrain at speeds up to 15 mph with ground and grade conditions requiring all-wheel drive.
h. Negotiating side slopes up to 20 percent on a surface reasonably hard and free from loose material.

3.7 Human factors engineering. The truck shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, 4 (general requirements), 5.1 (control/ display integration), 5.2 (visual displays), 5.3 (audio displays), 5.4 (controls) 5.5 (labeling), 5.8 (environment), 5.9 (design for maintainability), 5.12 (operational and maintenance ground/shipboard vehicles), and 5.13 (hazards and safety) of MIL-STD-1472, as applicable.

3.7.1 Noise limits. The noise produced by the truck shall conform to MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2 and 5.4, when tested in accordance with 4.5.2.7. The provision of MIL-STD-1474, 4.3 and 4.4, shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity, and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. If permission to exceed 85 dB(A) at the operator's position is obtained, then noise hazard signs shall be posted at the operator's position. The sign shall conform to MIL-STD-1474, 4.3, except the sign shall state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET" and shall be readable at the distance indicated on the sign. The distance shall be the maximum distance obtained from an 85 dB(A) contour conducted at the operator position as specified in 4.5.2.7.

3.8 Safety. All exposed parts that are subject to high operating temperatures or are energized electrically shall be insulated, fully enclosed, or guarded. All rotating, reciprocating, or moving parts that are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Fire fighting equipment, when mounted on the truck, shall be located and positioned not to present a safety hazard to personnel. Surfaces of the truck that are used as walkways, working areas, or steps, shall be provided with nonskid surfaces.

3.9 Environmental operation requirements.

3.9.1 Temperature. The truck and equipment shall be capable of starting and performing in any temperature from +125 °F to -25 °F.

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3.9.2 Altitude. The truck and equipment shall deliver its rated pumping capacity at 100-psi net pump pressure and 250 gpm at any elevation from sea level to 2000 feet at the temperature specified in 3.9.1. When specified (see 6.2), modification kits shall be provided and installed for use of trucks and equipment to specified elevations above 2000 feet.

3.10 Detailed requirements.

3.10.1 Engine. The truck shall be equipped with the manufacturer's current, standard, liquid-cooled, diesel engine that is rated for fire service use. The engine furnished shall produce the required vehicle performance when operated on fuel conforming to VV-F-800 at an engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of continuous, warranted operation. At the engine manufacturer's rated governed speed, the engine shall develop not less than 200 gross horsepower. Engine net torque ratings and new horsepower rating used in performance prediction shall be determined in accordance with SAE J245 or SAE J270, as applicable.

3.10.1.1 Brake horsepower certification. The contractor shall obtain a certified net brake horsepower curve for the engine being furnished and shall make the certification available for review by the contracting officer or designated representative.

3.10.1.2 Lubrication system. The engine lubrication system shall be the full-forced-feed type. A low-oil-pressure warning device shall be provided and located in the cab audio/visual. A full-flow type oil filter shall be furnished. An oil sampling valve conforming to Circle Seals Control's part number 59-500 or equal shall be installed in the lubrication lines. The valve shall be located to ensure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the valve shall be such that samples taken will be a true representation of oil that is flowing while the engine is running.

3.10.1.2.1 Drain plugs. All drain plugs installed throughout the truck for the purpose of draining lubricants shall be of the permanent magnet type.

3.10.1.3 Fuel system. Fuel system shall conform to DOT Motor Carrier Safety Regulations, section 393.65.

3.10.1.3.1 Fuel tank. The fuel tank shall be sized in accordance with NFPA No. 414 but shall not be less than 35 gal. Tank and fill piping shall be so placed as to be protected from mechanical injury. The tank shall be labeled "DIESEL FUEL ONLY", in accordance with MIL-STD-1472, 5.5.1.1 as specified in 3.7. Step tanks may be furnished, but side running boards are required as specified.

3.10.1.3.2 Fuel filter. The fuel system shall be provided with a fuel filter either installed at the tank or in the fuel line.

3.10.1.4 Engine starting system. The engine starting system shall prevent engine start unless the transmission is in neutral.

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3.10.1.5 Engine cooling system. The cooling system shall be standard for the truck furnished. Draincocks shall be installed at the low points of the cooling system and at such points as may be necessary to drain the system to prevent freezing. The cocks shall be of a hand-operated type that will remain closed under the most severe operating conditions to be encountered by the truck. A coolant temperature indicator shall be provided on the cab dash panel.

3.10.1.6 Engine exhaust system. The exhaust system shall conform to DOT Motor Carrier Safety Regulations, section 393.83. A spark arresting muffler or a spark arrester shall be provided. Muffler and tailpipe shall be made of aluminized steel. The exhaust system discharge shall be directed away from the pump operator's control panel area and tires. The muffler shall be located not closer than 2 ft from the fuel tank, with full shielding for the tank from any part of the exhaust system within 1 ft of the tank and insulation and shielding where needed elsewhere.

3.10.1.7 Air pollution control. The truck shall comply with the Federal EPA regulations governing control of air pollution from new motor vehicles and new motor vehicle engines in effect on the date of manufacture.

3.10.1.8 Governor. The governor shall be the manufacturer's current standard type for the engine. It shall be set and sealed to limit engine to maximum recommended operating speed.

3.10.1.9 Air cleaner. An air cleaner shall be furnished.

3.10.1.10 Engine hourmeter. An engine hourmeter of the oil pressure type, having a totalizing mechanism of not less than 9,999 hours, shall be furnished for the truck chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to ensure continuous trouble-free performance under severe operating conditions. The engine hourmeter shall be mounted on the truck instrument panel.

3.10.2 Truck drive assembly. The entire propulsion train between the engine and the drive wheels shall have a torque capacity at least equal to the maximum torque developed by the engine through the intervening speed and torque ratios.

3.10.2.1 Transmission. The vehicle shall be provided with a standard heavy-duty, manual, five-speed transmission. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

3.10.2.2 Transfer case. A two-speed type transfer case shall be provided. Unless the transfer case is equipped with devices that compensate for differential torque and speeds between front and rear axles, the transfer case shall provide driver selection of rear-axle or all-wheel drive. When furnished, interaxle compensating devices shall provide positive transfer of power to all driving axles.

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3.10.2.3 Axles. The front and rear axles shall have a rated capacity of not less than 110 percent of the load imposed. The rear axle shall be provided with positive traction or limited slip differential.

3.10.2.3.1 Drive hubs. Drive hubs for disengaging front wheels from the power train shall be provided for cab operation except for trucks equipped with interaxle compensating devices between front and rear axles.

3.10.3 Suspension. The truck shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, plus at least 10 percent, measured at the ground, with truck loaded to specified gvw. When suspension is rated at the spring pads, unsprung weight shall be deducted. The truck shall be equipped with hydraulic, heavy-duty, double-acting shock absorbers on the front. The chassis manufacturer's standard auxiliary rear springs shall be furnished.

3.10.4 Wheels, rims, tires and tubes. The truck shall have single front and dual rear wheels. Rim and tire ratings shall conform to Tire and Rim Association recommendations. Tire pressures recommended by the Tire and Rim Association Yearbook shall be stenciled in 1/2-inch-high letters above and near tires. These stencils shall read "TP (specify) PSI," in accordance with MIL-STD-1472, 5.5.1.1, as specified in 3.7. Tires shall be tube type of not less than 100-level quality and have standard commercial mud and snow tread. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations. All tires furnished on the truck shall be of the same size. The rated individual tire capacity at 60-mph highway conditions shall not be less than the maximum wheel loading imposed with truck loaded to rated gvw. Wheel and tire assemblies shall be dynamically balanced at simulated speeds up to 60 mph. Disk-type wheels shall be provided. Tire tracking front to rear shall be the same. Fender opening shall provide for duel snow chains on the rear wheels. Tires furnished to the Government shall be not more than 12 months old from date of cure of the rubber to date of acceptance by the Government.

3.10.5 Brakes. Certification shall be obtained from the brake manufacturer that the brakes are properly sized for the gvw of the truck. The certification shall be made available to the contracting officer or designated representative.

3.10.5.1 Service brakes. The truck shall be equipped with straight-air, all-wheel, self-energizing-type service brakes. Service brakes shall control and hold the truck, loaded as specified in 3.6, on a 50-percent grade. Under all conditions of loading, service brakes shall stop the truck within 30 ft from a speed of 20 miles per hour (mph) on dry, hard, level surfaces free of loose material. The brake mechanism shall be accessible for exterior adjustment. Brake lines shall be anchored to the chassis and protected from damage. Two methods shall be provided to prevent system freezing. All brakes shall conform to the latest DOT requirements.

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3.10.5.2 Parking brake. The truck shall be provided with a parking brake. The parking brake shall be the self-energizing, air-release type. The parking brake shall be controlled by a push-pull-type control on the cab dashboard. The air system when charged to not less than 100 psi shall not leak more than 25 percent in 24 hours.

3.10.5.3 Engine brake. The engine brake shall be an engine exhaust brake system designed for the engine used in the truck. Braking shall be accomplished by motion transfer through a master-slave piston arrangement which opens the cylinder exhaust valves near the top of the normal compression stroke, releasing the compressed cylinder charge to exhaust. The engine brake control shall be an "on-off" switch, mounted on the truck dash, that permits the engine brake system to be energized by removing pressure from the accelerator pedal when the switch is in the "on" position.

3.10.6 Steering mechanism. Provisions shall be made for rapid adjustment of lost motion in the steering mechanism. The steering wheel shall be of the standard commercial size, shall not interfere with any other projecting equipment or operation of the vehicle, and shall be mounted on a rigid and sturdy column. Front wheels shall have a cramping angle of not less than 28 degrees right and left at the wheel on the inside of the turning circle. The truck shall be equipped with a power-assisted steering mechanism of the hydraulic type. The mechanism shall preserve, to the maximum practicable extent, the features of directional stability inherent in the manually-operated steering mechanism, shall permit manual steering in the event that failure of the hydraulic system occurs with a force at the steering wheel complying with MIL-STD-1472, 5.4.2.2.5.7, and table IX, as specified in 3.7, and shall be effective at all vehicle speeds.

3.10.7 Front bumper and towing devices. A heavy-duty, full-width front bumper and two tow hooks shall be provided at the front of the truck. The two front tow hooks shall be mounted to the frame or rigid members attached to the frame. Two, 3-inch inside diameter, towing rings, or comparable U-bolts, shall be provided at the rear of the truck, and mounted directly to the chassis side rails.

3.10.8 Brush guard. A brush guard shall be provided. The brush guard shall be as wide as the widest part of the front fenders, as high as the hood in the center, and not less than 4 inches above the top of the front fenders. The brush guard shall be four-point suspension mounted.

3.10.9 Cab. The cab shall provide unobstructed seating for three people. The engine and driver's compartment shall be separated by an insulated firewall. The cab shall be upholstered and have a full-width adjustable seat. The cab shall be provided with boots for all levers and controls extending to the exterior of the cab. Safety grips shall be provided on each side of the cab to assist personnel in climbing into the truck cab. Fastenings, seatbelts, and armrests shall be furnished. Tinted safety glass shall be provided throughout the cab. Dual sun visors shall be furnished. A dome light shall be provided. The seat shall be upholstered over deep airfoam cushions with first grade Naugahyde, or equal, suitable for fire service.

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3.10.9.1 Windshield wiper and washer. Windshield wipers shall be mounted to facilitate clear vision. Wiper assemblies shall include features to permit speed control and return of the wiper blade to a retracted position out of the line of vision when not in use. The windshield cleaning system shall include a device that will spray the path of the wiper blades with water or cleansing solution. The cleansing solution reservoir shall have a minimum capacity of 2 quarts. The windshield wiping/cleaning system shall conform to SAE J198.

3.10.9.2 Heater and defroster. A hot-water heater shall be provided. The heater shall have fresh-air intakes. Discharge outlets shall be provided to direct heated air to floor and to defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver. The heater and defroster shall comply with 3.7.

3.10.9.3 Controls and operating mechanism. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from driver's compartment when dressed in fire fighting clothes. Instruments and controls shall be identified as to their function in accordance with 3.7 and installed in a manner to facilitate removal and servicing. Instruments shall be panel-mounted.

3.10.9.4 Stowage space. Stowage space of sufficient size to accommodate jack, handtools, antiskid chains, emergency reflective triangles, and reflectors shall be furnished for retaining equipment during truck operation. Storage space shall be labeled to indicate equipment stowed in accordance with MIL-STD-1472, 5.5.1.1, as specified in 3.7. Storage space for these tools may be furnished inside the cab.

3.10.10 Fire pump. The fire pump shall be the centrifugal type driven by an auxiliary diesel engine. The pump impeller shaft shall be of corrosion-resistant steel. The impeller suction inlets and all other portions of the pump coming into contact with the water shall be resistant to the corrosive effect of salt water. All pump-shaft glands or mechanical seals shall be adequate for service under operating conditions specified herein. The pump shall withstand a hydrostatic pressure of 350 psig for a period of 10 minutes. Draincocks, push-pull or rotary-turn type-accessible at the pump operator's position and labeled so that the open and closed position is clearly indicated, shall be provided to drain the pump and all water-carrying lines. Draincocks shall be of the water-service type and shall be equipped with positive stops. Pump drain lines shall be furnished and shall completely drain and discharge to a point below the bottom of the chassis frame. All threaded connections shall be NPT and shall conform to FED-STD-H28. The pump, when drafting at a minimum lift of 10 ft through 20 ft of 3-inch suction hose, shall:

a. Deliver 100-percent rated capacity at 150-psig discharge pressure at less than the manufacturer's recommended governed speed of the engine.
b. Deliver 70 percent of rated flow at 200-psig discharge pressure at less than the manufacturer's recommended governed speed of the engine.

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c. Deliver 50 percent of rated flow at 250 psig net pump pressure.
d. Deliver each flow rate specified above at an additional 10-percent discharge gage pressure.
e. The pump, when dry, shall take suction and discharge water within 30 seconds, with a lift of 10 ft through 20 ft of 3-inch suction hose.
f. The pumping requirements specified herein shall be accomplished at all altitudes up to 2000 feet above sea level and in any ambient temperatures from +125 °F to -25 °F without evidence of overheating, vapor lock, excessive vibration or any other malfunction. When operation above 2000-feet altitude (see 6.2) is required, special provisions may be made.
g. The fire pump shall deliver not less than 250 gpm from the water tank and shall maintain the rate of flow for not less than 80 percent of the rated capacity of the tank.

3.10.10.1 Auxiliary engine. The fire pump shall be furnished with an electric-start, air-cooled diesel engine. The engine shall be selected in accordance with MIL-STD-1410, 1.2.1 using the appropriate class and shall have sufficient power to meet the pump requirements of 3.10.10. The engine shall start from the main truck battery. A start button shall be furnished on the pump panel and in the cab accessible to the driver. The bottom shall be labeled "PUMP ENGINE START", in accordance with MIL-STD-1472 5.5 as specified in 3.7. Fuel for the engine shall be supplied from the main truck fuel tank. The pump and engine shall be mounted as a single unit at the rear of the truck on the same platform as the booster reel.

3.10.10.1.1 Priming device. The priming device shall be provided, complete with all necessary valves, with levers operated from the pump control panel. The priming device shall develop a vacuum of 22 inches of mercury (Hg) at an altitude up to 1000 ft above sea level.

3.10.10.2 Suction connections. The fire truck shall be provided with two gated, 2-1/2-inch, full-flow type suction connections, one located to take water from the tank and one to take water from an auxiliary source. The valves shall be the pump manufacturer's standard corrosion-resistant, balanced-ball type for quarter-turn operation. The valves shall be operated and locked by means of an individual control handle as a single-handed operation from the operator's panel. Valve controls shall be provided with double-ended arrows showing the direction of operations and labeled at each end to indicate the functional result (e.g. open and close), in accordance with MIL-STD-1472, 5.4.1.2.4, as specified in 3.7. The auxiliary valves shall be fitted with a female swivel and a rocker-lug plug. The suction connections shall be equipped with a removable and accessible strainer; shall have a machined end with mirror finish to permit hand-tight connections; shall provide a seat for suction hose gaskets; shall be labeled "INLET" in accordance with MIL-STD-1472, 5.5.1.1, as specified in 3.7; and shall be threaded with American National firehose coupling thread (NH). All piping from the suction connection shall be 3-inch inside diameter. The suction connection for auxiliary sources shall be used to draft water from natural sources through 20 feet of 3-inch suction hose with a vertical height of 10 feet. Two 10-foot sections of UL approved suction hose shall be supplied with each

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truck. The hose shall be stowed in two suction-hose trays or tubes. The trays or tubes shall be installed on the truck to allow personnel standing on the ground to easily remove the hose. The trays or tubes shall be equipped with positive-locking mechanical devices to prevent hose movement under all road conditions. The trays or tubes shall be designed to provide water drainage over their full length. Finish shall be such as to prevent hose chafe.

3.10.10.3 Discharge connections. The fire truck shall be provided with one 2-1/2-inch discharge connection located on the pump operator's control panel, two 1-1/2-inch discharge connections, one in each of the 1-1/2-inch transverse hosebeds, and one discharge connection to the booster reel. All discharge connections shall have machined ends with a mirror finish to permit hand-tight connections. All discharge connections shall be labeled "OUTLET", in accordance with MIL-STD-1472, 5.5.1.1, as specified in 3.7. The connection shall provide a seat for discharge hose gaskets, and shall be threaded with NH threads. The 2-1/2-inch connection shall be fitted with a 2-1/2 by 1-1/2 NH adapter with 1-1/2-inch gasketed cap and chain. The discharge connections shall be provided with a single discharge pressure gage located on the fire pump operator's control panel. The discharge valves shall be the pump manufacturer's standard valve, and shall be corrosion-resistant, ball-type for quarter-turn operation. The 2-1/2-inch valves shall be operated and locked by means of an individual control handle as a single-handed operation Akrow Remote Control Style 1488 or equal. Discharge connections shall be provided with drain valves of at least 3/4-inch inside dia for draining hose lines with the pump discharge gates closed. Push-pull or rotary-turn type drain valves shall be of the water service type and shall be equipped with positive stops. Drain-valve controls shall be located adjacent to the discharge connections. Drain lines shall be concealed behind the pump panels and shall discharge to a point below the bottom of the chassis frame. Each discharge connection, pressure gage, and drain shall be identified by a related numbering system as well as function.

3.10.10.3.1 Dump valve discharge. Water pipe not less than 4 inches in diameter shall be piped to the rear of the truck. The water pipe shall be used for rapid dumping of the water tank when the fire truck is used as a tanker. A discharge line from the pump may be piped into the center of the dump line to assist in rapid dumping of the water tank. The dump line and the discharge line shall be furnished with control valves, and shall be controlled from the rear of the truck by personnel standing on the ground. The dump valve shall be of the butterfly type with a positive locking one-handed control handle. The dump line shall hace 4-1/2-inch National hose (NH) male threads and machined ends for suction hose coupling or valve gasket seating. The end shall be furnished with a readily accessible removable brass strainer and cap.

3.10.10.4 Relief valve. A relief valve may be provided.

3.10.10.5 Pressure rating. The pump shall withstand not less than 350-psig hydrostatic pressure on the discharge side for 10 minutes.

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3.10.10.6 Fire pump operator's control panel. The fire truck shall be equipped with a stainless steel fire pump operator's control panel located at the rear of the truck. The panel shall have a nonglare finish. The pump pressure and suction gages shall not be less than 3-1/2-inch dia liquid-filled or single-Bourdon type, with throttling valves or restrictive devices to eliminate pulsating of gages. The pump pressure and suction gages shall be accessible to the operator. All gages shall read zero when the equipment is shutdown. All valve operating handles and other controls shall be operable by personnel wearing heavy gloves. All extended controls and handles shall be noninterfering and be firmly supported. The panel shall include at least the following:

a. Pump pressure gage - 30 inch Hg (vacuum) to 600 psi.
b. Pump suction gage - 30 inch Hg (vacuum) to 600 psi.
c. All pump discharge and suction valve controls.
d. Tank suction valve control and instructions.
e. Primer device control and instructions.
f. Illumination of all instruments and controls with nonglare lighting.
g. Engine start and stop control.
h. Water tank level gages, electronic type, shall be installed in the cab and on the pump panel.
i. All other pump-panel-mounted drains and controls mentioned herein.
j. Discharge valve and throttle handles shall be of black, extreme-temperature-resisting plastic, brass, or stainless steel. All gage rims and other exposed metal shall be chromium plated or stainless steel.
k. A discharge pressure gage shall be installed in the cab for use during pump and roll. The gage shall be equipped with a throttling device to eliminate pulsating and shall be not less than 3-1/2 inches dia. The gage shall be visible to both the driver and assistant driver.
l. Instruction plates specified in 3.15.
m. A oil pressure gage for the pump engine shall be installed on the pump panel with a audio and visual warning low oil pressure device mounted with the water tank level gage located in the cab.

All controls, valve drains, and gages shall be identified by function and shall have a related identification number. All instruments and switches shall be identified by function as well. Valves controls shall be provided with double-ended arrows showing the direction of operations and labeled at each end to indicate the functional result (e.g. open and close), in accordance with MIL-STD-1472, 5.4.1.2.4, as specified in 3.7. The control panel shall comply with 3.7.

3.10.11 Water tank. The water tank shall be fabricated of 304 L, high-strength, corrosion-resistant, stainless steel that has been passivated in accordance with ASTM A 380, and shall be free of mill scale either by a pickling treatment or sandblasting.

3.10.11.1 Tank filler opening. The tank shall have an accessible filler opening not less than 144 square inches and shall be provided with a tight, gasketed, hinged cap having a holddown catch. The tank fill opening shall be

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provided with a removable strainer. A 2-1/2-inch line from the pump to the tank shall be provided for filling the tank, with a valve control located on the pump panel.

3.10.11.2 Tank mounting. The tank shall be supported on sills to cushion shock and transfer the load to the truck frame. The tank shall be so mounted to the chassis that it can be removed.

3.10.11.3 Tank piping. All piping shall be of corrosion-resistant metal or shall be galvanized. The pipeline may incorporate flexible, reinforced, noncollapsible connection to absorb vibration. A tank-to-pump valve shall be provided with controls located on the pump operator panel and to the cab tank fill gage. The valve shall be provided with a device that will not allow the pump engine to start from the cab unless the tank-to-pump valve is open. The tank piping shall permit discharging tank water through the fire pump discharge manifold at a flow rate of not less than 250 gpm. The tank suction pipeline, valves, and fittings shall withstand not less than 120-psi working pressure. The piping between the pump, booster reel, and transverse hose-beds shall be not less than 1-1/2-inch inside dia and shall incorporate 1-1/2-inch, quick-opening valves with controls extended to the pump operator's panel. The discharge piping, valves, and fitting between the pump booster reel and transverse hosebed shall withstand not less than 500-psi working pressure. If manuals are furnished with the fire truck a complete piping diagram shall be provided. All drain lines shall discharge to a point below the bottom of the chassis frame. The piping system shall provide the capability of removing 80 percent of tank capacity when operating upgrade or downgrade on a 30-percent grade.

3.10.11.4 Tank-level gage. A water-tank-level gage shall be furnished and located on the fire pump operator's control panel. The gage shall be of the electronic type. Another water-tank-level gage shall be installed in the truck cab visible to the driver and passenger. The gage shall be mounted on a panel with all other controls and devices specified herein to control or monitor the pump from the cab.

3.10.12 Body. The truck shall be equipped with a body that incorporates a fire pump housing and stowage bin, fire pump operator's control panel, removable waterhose bed, equipment cabinets, rear step, booster hosereel, and fire fighting equipment. The body shall be made of high strength material and shall be of welded construction. The body shall be reinforced at all points where clamps and hangers are attached. The body shall be free of rough and sharp edges. The body shall be furnished with flexible mud-flaps and fenderetts to reduce tire splash in wet weather.

3.10.12.1 Hosebed. A hosebed shall be provided with the capability of stowing not less than 400 ft of 1-1/2-inch, double-jacketed, cotton, rubber-lined, firehose. The hosebed interior shall be free of all projections and sharp or rough edges that might chafe or tear the firehose or hinder its placement or removal. The hosebed sides and end shall be flanged and reinforced to withstand not less than a 200-lb horizontal pull at any point on the top edge without permanent deformation. The hosebed floor shall be of slatted aluminum or well-

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seasoned, select-grade oak or other hardwood having equal characteristics or plastic-base space age state-of-the art material. The slats shall run the full length of the hosebed, and shall be spaced to provide ventilation for the hose. The hosebed floor assembly shall be removable. The interior of the hosebed shall be free of all projecting nuts, sharp angles, or brackets. Exposed bolt or rivet heads shall be of the oval type. Two folding, spring-loaded, chromium-plated toe steps shall be furnished. Toe steps shall be sized in accordance with MIL-STD-1472, 5.6.3, as specified in 3.7. One shall be located on the right and one on the left at the rear of the water tank or body to permit easy access to the hosebed.

3.10.12.2 Transverse hosebeds. Two transverse hosebeds shall be located over the top of the pump area. Each transverse bed shall have the capability of stowing not less than 200 ft of 1-1/2-inch, double-jacketed, cotton, rubber-lined, firehose. Hose playout or removal openings shall be provided on each side of the truck with hose rollers installed in the bottom and both sides of each transverse bed opening to prevent hose interference or snagging. Two 1-1/2-inch, swivel-type discharge connections shall be provided to allow hose to be removed from either side of the truck. The bottom of the transverse hosebeds shall be of the same material as used in 3.10.12.1. Discharge gate controls for each hosebed shall be located at the pump control panel and shall be suitably identified. The hosebeds shall be identified respectively as transverse hosebed No. 1 and 2. Hosebed No. 1 shall be located toward the front of the vehicle. Each hosebed shall be provided with a center-mount, swivel-type discharge connection to permit hose to be removed from either side of the truck. The discharge connections shall be located away from any hosebed to permit hose attachment to, or removal from, the connections without sidewall interference.

3.10.12.3 Equipment cabinets. All equipment cabinets and cabinet doors shall be the manufacturer's standard box-type construction and shall be watertight. Cabinets shall be framed of welded- or bolted-in structural members for strength and resistance to damage. Cabinet doors shall be of double-panel construction and attached with full-length, corrosion-resistant, piano hinges. All vertically hinged doors shall have automatically engaging, spring-type, hold-open devices. Each cabinet door shall be provided with recessed "D-ring" type chromium-plated handles, operable with heavy-gloved hands, with weather-resistant latches. Maximum corrosion-resistant materials shall be used in cabinet fabrication and hardware. Doors shall be fully gasketed and weather-stripped with neoprene and all cabinet seams shall be complete welds. Drip rails or flanged cover-plates shall be provided to keep rainwater from entering cabinets. Provisions shall be incorporated to permit flushing and cleaning inside of cabinet. Each cabinet shall be illuminated. Cabinet lights shall be controlled automatically by individual door switch, with an indicating light, located on the truck instrument panel. Cabinet tops and wheel-housing tops shall run continuously on each side of the truck and shall be of antiskid surface. All equipment cabinets specified herein shall be a minimum of 1/8-inch aluminum with 3/16-inch aluminum doors. Cabinets shall not be flush with the wheel skirting. Cabinet bottoms shall be fitted with the same material used in the hosebeds. Cabinet door latches shall meet KKK specification for patient carrying vehicles. Drain holes shall be provided.

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3.10.12.3.1 Cabinet arrangement. The streetside and curbside cabinets shall be identical and shall both run the full length of the water tank. One cabinet on each side shall be located aft of the rear wheels. Two cabinets shall be located ahead of the rear wheels. The cabinets shall be the maximum size possible. The cabinets adjoining the rear wheels shall have vertically hinged doors with the hinges located on the door edge toward the front of the truck. Each forwardmost cabinet on either side shall have dropdown doors and be sized to accommodate the stowage of not less than five fire rakes, five long-handled shovels, and five forestry McLeod tools. However, these tools shall not be furnished with the truck.

3.10.12.3.2 Hosereel compartment. A rear hosereel compartment shall be furnished. The hosereel specified in 3.10.18 shall be mounted at the rear of the truck. Access to the hosereel shall be from the rear of the truck. There shall be sufficient width to permit access to the hosereel for servicing and removal and hose playout. The hosereel shall be illuminated and the lighting shall be controlled by the switch specified in 3.10.12.3. Hose rollers shall be installed on both sides and bottom of the hosereel to prevent hose interference or snagging. The hosereel shall be mounted on a common platform with the auxiliary engine and pump. Mounting shall be to the truck frame or frame extensions. The compartment shall be provided with a roll-up door.

3.10.13 Electrical equipment. The truck shall be provided with a complete 12-volt, negative-ground starting and lighting system. Truck lights and wiring shall conform to DOT Motor Carrier Safety Regulations, sections 393.12, 393.19, 393.20, 393.23, 393.24, 393.25, 393.27 through 393.31, and 393.33, except as specified herein. Reflectors shall conform to DOT Motor Carrier Safety Regulations, section 393.26. Wiring shall be weatherproof, resistant to moisture and fungus, and color-coded or numbered for ready identification. Wiring identification shall be consistent throughout the truck. Two spare, single-pole, 15-ampere (A) breakers shall be furnished for future radio use. The truck chassis electrical circuits and the truck body electrical circuits shall be separate circuits. The truck body electrical circuits, excluding truck chassis electrical circuits, shall be provided with overload protective devices of the manual or automatic reset, overload-circuit-breaker type. All electrical equipment shall conform to MIL-STD-454, requirement 1.

3.10.13.1 Battery system. A dual 12-volt (V) battery system shall be provided. Each battery shall be 12 V and of the sealed-cell (maintenance-free) type. Each battery shall have a cold crank rating of at least 450 amps and a reserve capacity rating of at least 125 minimum, as set forth in SAE J537. The battery units may be mounted on a sliding tray in an enclosed compartment and shall be accessible for testing and servicing. A pictorial battery connection diagram shall be provided that is readable when batteries are being serviced. The batteries shall be securely held in place on the trays with removable locking devices and shall be protected against engine and exhaust heat and road splash. Provisions shall be made for securely locking the sliding trays in the retracted position. The battery compartments shall be ventilated and treated against acid fumes and spillage. The battery compartments shall not be mounted on or project onto the side steps when in the retracted (operating) position. Batteries shall

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be provided with polarized receptacles for station charging. The receptacles shall be located at the rear of the truck in a manner to disconnect without damage when the truck moves forward. Each receptacle shall be the recessed female type with a spring-loaded cover and provided with matching male plug protected prongs. The receptacle shall be labeled "CHARGE ACCESS" in accordance with MIL-STD-1472 5.5.1.1, as specified in 3.7. A carrying handle for the batteries shall be provided. Removable battery compartment covers shall be fitted with lifting handles; finger-lift openings are not acceptable. The battery terminals shall be provided with guards to prevent accidental shorting. Battery cables shall be colored black and labeled "-" for negative and colored red and labeled "+" for positive, in accordance with MIL-STD-1472, 5.12.9.1.4, as specified in 3.7. Battery circuits and switching system shall be provided whereby:

a. Each battery may be used:

(1) For starting alone.
(2) For starting and lighting.
(3) For operation of priming pump.
(4) For operation of electric booster hose reel.

b. Both batteries may be used simultaneously in starting.

3.10.13.1.1 Electric generating system. The engine shall be equipped with a 12-V alternator and regulator capable of supplying the built-in direct current (dc) requirements of the truck with a minimum rating of 145 amps. Charging rate at 1000 engine revolutions per minute (rpm) shall be a minimum of 70 amps.

3.10.13.2 Lighting system. All truck lights, reflectors, and wiring shall conform to DOT Motor Carrier Safety Regulations, except as specified herein:

a. DOT identification lamps are required.
b. Clearance and side-marker lamps shall be combined and mounted to indicate width of trucks.
c. Overload protective device shall be of the automatic-reset, overload-circuit-breaker type. The lighting system shall be arranged so that the headlamp circuit or circuits shall not be affected by a short circuit in any of the other circuits.
d. Rear-mounted lights shall be located so that they shall not be obscured by equipment or personnel normally standing on the rear step.
e. Front-turn-signal units shall be standard commercial type. The self-canceling-type controller shall be mounted on the steering column, shall operate a visual indicator, and shall be wired through the ignition system.

3.10.13.2.1 Additional lighting. The following additional lighting shall be provided:

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a. Two automotive-type sealed-beam spotlights located on each side of the windshield, with controls extended inside the cab for "on-off" control and movement of the lamp housing through horizontal and vertical angles. Spotlight control handle shall not obscure the driver's vision.
b. Two sealed-beam hose pickup lights. Lights shall be mounted to the outside of the hosebed.
c. Shielded engine-compartment lights arranged to illuminate both sides of the engine. Lights shall be controlled by a switch installed in the cab.
d. Pump control panel lights enclosed within hoods or covers designed to completely illuminate the panel controls, pump, motor, and hosereel.
e. Two backup lights, with not less than 5-inch diameter, located at rear of the body. The backup light switch shall be automatically controlled when the transmission is placed in the reverse mode.
f. Stoplights, taillights, rear-turn signal lights, and backup lights shall be flush mounted at the rear of the body.
g. A complete wiring diagram shall be provided with each truck, except when provided in a manual furnished with the truck.

3.10.13.2.2 Warning lights. Warning lights shall conform to NFPA No. 1901. Warning lights shall be as follows:

a. One rotating, three-unit, sealed-beam lamp, mounted centric on top of the truck cab, with brush guard.
b. Two strobe-type warning lights mounted on either side of the rear of the truck outside the hose pickup lights.
c. Two low-mounted, 6-inch diameter, flashing lights on the front of the truck, installed on a horizontal plane through the centerline of the headlights. Lights shall not be mounted on the brush guard.

All warning lights shall have red lenses. The controls for the warning lights shall be independent of other electrical equipment controls and shall be broken down into three groups: labelled front, top, and rear. Warning lights shall be 12 Vdc.

3.10.13.2.3 Light switches. The headlight switch shall be the standard push-pull type. The warning light shall be the single-pole-single-throw, rocker, circuit-breaker type with indicating light for the "on" position. There shall be one switch for each of the groups specified in 3.10.13.2.2. All switches shall be clearly and permanently marked. Indicating lights shall be shielded to prevent glare on the windshield during night driving.

3.10.13.3 Siren and horn. The truck shall be equipped with a Federal Sign and Signal Corporation electronic siren and PA combination Model BA20A or equal. The electronic siren shall be furnished with a microphone and all wiring applicable to a two-way radio. A single 6-inch diameter speaker shall be mounted behind the truck grill. Controls for the siren shall be mounted for easy access for the driver and passenger.

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3.10.13.4 Backup warning device. The fire truck shall be provided with an automatically activated, audible, pulsating, signaling warning device, electrically operated to caution personnel when the fire truck is in reverse gear operation. The backup warning device shall conform to SAE J944.

3.10.14 Rear platform. The truck shall be provided with a rear platform. The rear platform shall have a load-carrying capacity of not less than 500 lbs. The rear platform length shall be equal to the width of the truck. The platform shall be not less than 12 inches in width, from front to rear, in front of the hosereel compartment. Rear platform shall have rounded corners and shall be flanged downward at least 2 inches. The rear platform shall have a nonslip tread surface. The rear platform shall be 22 inches +2 inches -0 inch, above ground level, with the truck loaded as specified in 3.4.2.1. The rear platform shall be the fold-up type to allow for off road operation and is not intended to meet angle of approach and departure requirements. The platform shall be designed to be securely locked in the up position under all conditions of operation.

3.10.15 Handrails and grabrails. The rear platform shall be provided with two vertical grabrails to assist personnel in climbing onto the rear platform. A horizontal handrail shall be provided at the rear platform and not more than 66 inches above the rear platform. The handrails and grabrails shall be 1-1/4 inch minimum diameter chromium-plated brass, chromium-plated steel, or polished stainless steel tubing. All handrails and grabrails shall be of the nontwist type.

3.10.16 Steps. Steps shall be provided at the rear and sides of truck to assist personnel in gaining access to fire fighting components.

3.10.17 Buzzer. A buzzer shall be installed in the cab and a weatherproof pushbutton on one of the rear handrail supports to provide a signaling device from the rear platform to the truck cab. The buzzer's signal shall be at least 20 dB above the typical operating noise levels, in accordance with MIL-STD-1472, 5.3.4.1, as specified in 3.7. The push button shall be located so personnel standing on the ground can operate the button.

3.10.18 Booster hosereel. A booster hosereel shall be provided and installed as specified in 3.10.12.3.2. The reel shall be an electrical-rewind type. The rewind control switch shall be a push type, spring-loaded button. The electric-rewind control switch shall be installed near the hosereel. The switch shall be suitably identified. The hosereel shall be equipped with 150 ft, in three 50-ft lengths, of 1-inch, 800-psi, rubber chemical hose, jointed with smooth, full-flow type, 2300-psi couplings. A 1-inch brass hose nozzle of the combination straight-stream and adjustable fog type shall be furnished. The nozzle shall be equipped with ball shutoff and click-stop nozzle settings. The nozzle shall be of the pistol-grip type. Nozzle tip shall be fitted with a rubber or synthetic rubber-ring. The nozzle shall permit preselection of any nozzle setting prior to turning on the nozzle as well as shutoff at any nozzle setting. The nozzle shall provide settings for straight-stream and minimum fog settings of 30, 60, and 90 degrees. The nozzle shall have a discharge capacity of not less than 20 gpm for straight stream and not less than 22 gpm for fog (spray). The hosereel shall

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have a friction brake playout to prevent overtravel. The hosereel shall also have provisions for manual handcrank rewind, with the handcrank facing toward the firefighter standing at the rear of the truck. The hosereel shall be connected to the discharge manifold of the fire pump with a 1-1/2-inch inside dia galvanized steel pipeline having a flexible section to absorb vibration, and shall have an operation valve with handle extended to the left side of the truck on the pump operator's panel. The discharge pipeline and fittings between the pump and booster reel shall withstand not less than 500-psi working pressure. The drain line shall discharge at a point below the bottom of the chassis frame.

3.10.19 Communication equipment. A radio communication system space for installation of communication equipment approximately 1-1/2 ft long by 1-1/2 ft wide by 1 ft high shall be provided.

3.10.19.1 Radio noise suppressors. The truck chassis shall be equipped with radio noise suppressors conforming to SAE J552. Electromagnetic radiation from the truck chassis shall be within the limits of SAE J551.

3.10.20 Servicing and adjusting. Prior to acceptance of the truck by the Government inspector, contractor shall service and adjust each truck and its mounted equipment for operational use, including at least the following: focusing of lights; adjustment of engine; adjustment of electrical and brake system; alignment of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear and mounted equipment; servicing of cooling system with a solution of antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

3.10.21 Rustproofing. The truck shall be rustproofed in accordance with MIL-STD-1223.

3.11 Physical protection.

3.11.1 Lubricants. The procedure for the selection of lubricants shall be in accordance with section 5 of MIL-STD-838. Lubricants selected shall be in accordance with chapter 2 of MIL-HDBK-113. When the specification of the lubricant selected includes a requirement for a Qualified Products List (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.11.1.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected, accessible position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a hydraulic coupler, conforming to MS24207, and include a 10-inch flexible extension. Accessibility to lubrication fittings shall be provided without removal or adjustment of accessories or parts. Panels or plates equipped with hand-operated fasteners may be removed to provide accessibility. Accessibility shall be in accordance with SAE J925 for personnel normally clothed.

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3.11.1.2 Enclosures with integral reservoir of lubricant. Enclosures such as engine crankcase, transmission housing, and gear cases which contain a reservoir of liquid lubricant shall be equipped with dipsticks marked in accordance with SAE J614, check plugs not less than 0.5-inch pipe size, or sight gages to determine the acceptance level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining the lubricant. Drains shall be fitted with an SAE J531 or J532 magnetic drainplug. The drainplug shall be located so that removal of the plug will result in complete drainage of the lubricant. Drainage shall be to a container on the ground when the truck is on level ground. Integral tubes or troughs may be used to convey the lubricant from the drain to the container. Accessibility to the drainplug, the filling means, and the lubricant-level checking device shall be provided without removal or adjustment of accessories or parts. Panels or plates equipped with hand-operated fasteners may be removed to provide accessibility. Accessibility shall be in accordance with SAE J925 for personnel normally clothed.

3.11.2 Fungus- and moisture-resistance. Electrical circuitry, including all components and connections, except as specified below, shall be protected from the effect of fungus growth and moisture penetration by an overall treatment with a varnish conforming to MIL-V-173, composition I:

a. Components or circuit elements which are inherently fungus- and-moisture resistant, or which are hermetically sealed need not be treated.
b. Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to provide a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that surfaces are free of all foreign matter that would interfere with the adherence or function of the varnish.

3.11.2.1 Varnish composition. When local air-pollution regulations preclude the use of MIL-V-173, composition I varnish, composition II varnish may be used for fungus and moisture- resistance. When composition II varnish is used, the contractor shall have evidence on hand that the use of composition I varnish is precluded and that the composition II varnish complies with Rule 442 of the South Coast Air Quality Management Control District.

3.12 Electromagnetic interference (EMI). Unless otherwise specified (see 6.2), the EMI emission and susceptibility characteristics of the truck shall be in accordance with SAE J551.

3.13 Color and finish. Unless otherwise specified (see 6.2), the truck exterior shall be cleaned treated and painted in accordance with MIL-T-704, type A, TT-E-489, or commercial equivalent. Those parts of the truck normally painted shall be gloss line-yellow conforming to FED-STD-595, color number 13670, except

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the truck cab roof down to the bottom of the front windshield shall be painted gloss white. All walk and step areas shall be polished anodized aluminum. Undercarriage and underbody shall be in accordance with MIL-STD-1223.

3.14 Identification marking. The truck shall be identified in accordance with MIL-STD-130. The marking shall be applied to the truck on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets to the inside of the door on the driver's side of the cab. The following data shall appear in the ownership and nomenclature spaces provided on the identification plate in their respective order:

OWNERSHIP - U.S. ARMY

NOMENCLATURE - TRUCK, FIRE FIGHTING: BRUSH, TANK,

6 X 6, COMMERCIAL CHASSIS

3.15 Instruction plates. All pumping controls, valves, instruments, and truck operating controls shall be provided with instruction plates designating function and operation. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material, size as applicable. Each plate shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.16 Contractor-furnished components. The following equipment components shall be furnished, installed, and stowed aboard each truck:

a. One 12-foot, aluminum, roof-type ladder conforming to NFPA 1961.
b. One combination adjustable wrench for hydrant and spanner.
c. One portable, floating-type, gasoline-engine-driven, self-priming pump, with a rating of not less than 120 gpm. The pump shall include all equipment components required for line service use, except hose and nozzle.
d. One chain saw conforming to GGG-S-62, type II, style 1, class B, with repair and maintenance kit.

3.17 Workmanship.

3.17.1 Castings and forgings. All parts, components, and assemblies of the truck, which include castings and forgings, shall be free of harmful extraneous material such as sand, dirt,
sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.17.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of the material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing. All bends shall be made by controlled means to ensure

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uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges and corners, except where sharp edges of corners are required or where they are not detrimental to safety.

3.17.3 Bolted connections. Boltholes shall be accurately formed and burrs shall be removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full-thread engagement. Bolts shall protrude not less than two full threads and not more than 5 full threads.

3.17.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, in full contact with the surface of the member, and shall be in accordance with SAE J492 or Industrial Fastener Institute Standard No. 114 for break mandrel blind rivets.

3.17.5 Welders and welding.

3.17.5.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME	- Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1	- Structural Welding Code, Section 5, Qualification.

Certification shall be made available for review by the contracting officer or designated representative. Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME or ANSI/AWS codes.

3.17.5.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale, and other foreign matter that can be removed by chipping and wire brushing. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the truck to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the truck shall be at the option of the contractor.

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3.17.5.3 Aluminum welding. The aluminum surfaces of parts to be welded shall be brushed with a stainless steel brush until free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, and lack of prescribed fit or splatter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article test. The first article shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The truck shall be examined as specified herein for the following defects:

Number		Requirement Defectparagraph
Major		
101.	Materials not as specified.	3.3
102.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments.	3.3.1
103.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1
104.	Contractor does not have documentation available for identification of materials, material finishes, or treatment.	3.3.1.2
105.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the truck.	3.3.2
106.	Truck chassis not as specified.	3.4.1
107.	The truck manufacturer's gross weight (gvw) rating for the truck chassis not as specified.	3.4.2, 3.4.2.1
108.	Dimensions not as specified.	3.4.2.2
109.	The chassis frame not as specified.	3.4.3
110.	Angles of approach and departure not as specified.	3.4.3.1
111.	Truck, components and accessories not as specified.	3.5.1
112.	Interchangeabilty of parts having the same part number not as specified.	3.5.2
113.	Maintenance support assemblages or elements not adequate.	3.5.3.1
114.	Ease of maintenance not as specified.	3.5.3.2
115.	Human factors not as specified.	3.7

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Number		Requirement Defectparagraph
Major		
116.	Noise level not as specified.	3.7.1
117.	Safety not as specified.	3.8
118.	Engine not as specified.	3.10.1
119.	Brake horsepower certification not furnished by contractor as specified.	3.10.1.1
120.	Lubrication of the engine not as specified.	3.10.1.2
121.	Drain plugs not as specified.	3.10.1.2.1
122.	Engine fuel system not as specified.	3.10.1.3
123.	Fuel tank(s) not as specified.	3.10.1.3.1
124.	Fuel filter not installed.	3.10.1.3.2
125.	Starting system not as specified.	3.10.1.4
126.	Engine coolant system not as specified.	3.10.1.5
127.	Exhaust system not as specified.	3.10.1.6
128.	Air pollution control not as specified.	3.10.1.7
129.	Engine governor not as specified.	3.10.1.8
130.	Air cleaner not furnished as specified.	3.10.1.9
131.	Engine hourmeter not as specified.	3.10.1.10
132.	Truck drive assembly not as specified.	3.10.2
133.	Transmission not as specified.	3.10.2.1
134.	Transfer case not as specified.	3.10.2.2
135.	Drive axles not as specified.	3.10.2.3
136.	Drive hubs not as specified.	3.10.2.3.1
137.	Suspension system not as specified.	3.10.3
138.	Wheels, rims, tires and tubes not as specified.	3.10.4
139.	Brakes not as specified.	3.10.5
140.	Service brake not as specified.	3.10.5.1
141.	Parking brake not as specified.	3.10.5.2
142.	Parking brake not as specified.	3.10.5.3
143.	Steering mechanism not as specified.	3.10.6
144.	Front bumper and towing device not as specified.	3.10.7
145.	Brush guard not as specified.	3.10.8
146.	Cab not as specified.	3.10.9
147.	Windshield wiper and washer not furnished as specified.	3.10.9.1
148.	Heater and defroster not as specified.	3.10.9.2
149.	Controls and operating mechanism not as specified.	3.10.9.3
150.	Stowage space not as specified.	3.10.9.4
151.	Fire pump not as specified.	3.10.10
152.	Auxiliary engine not as specified.	3.10.10.1
153.	Priming device not furnished as specified.	3.10.10.1.1

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Number		Requirement Defectparagraph
Major		
154.	Suction connections not furnished as specified.	3.10.10.2
155.	Discharge connections not furnished as specified.	3.10.10.3
156.	Relief valve provided.	3.10.10.4
157.	Controls not as specified.	3.10.10.6
158.	Water tank not as specified.	3.10.11
159.	Tank filler opening not as specified.	3.10.11.1
160.	Tank mounting not as specified.	3.10.11.2
161.	Tank piping not as specified.	3.10.11.3
162.	Tank level gage not as specified.	3.10.11.4
163.	Body not as specified.	3.10.12
164.	Hosebed not as specified.	3.10.12.1
165.	Transverse hosebed not as specified.	3.10.12.2
166.	Equipment compartment not as specified.	3.10.12.3
167.	Hosereel compartment not furnished as specified.	3.10.12.3.2
168.	Electrical equipment not as specified.	3.10.13
169.	Battery units and electrical circuits not as specified.	3.10.13.1
170.	Electrical generating system not as specified.	3.10.13.1.1
171.	Lighting system not as specified.	3.10.13.2
172.	Additional lighting not as specified.	3.10.13.2.1
173.	Warning lights not as specified.	3.10.13.2.2
174.	Light switches not as specified.	3.10.13.2.3
175.	Siren and horn not as specified.	3.10.13.3
176.	Backup warning device not as specified.	3.10.13.4
177.	Rear platform not as specified.	3.10.14
178.	Handrail and grabrails not as specified.	3.10.15
179.	Steps not as specified.	3.10.16
180.	Buzzer not as specified.	3.10.17
181.	Booster hosereel not as specified.	3.10.18
182.	Communication equipment not furnished as specified.	3.10.19
183.	Truck chassis not equipped with radio noise suppressors as specified.	3.10.19.1
184.	Truck not serviced and adjusted as specified.	3.10.20
185.	Truck not rustproofed as specified.	3.10.21
186.	Lubricants not as specified.	3.11.1
187.	Lubrication fittings not of type, location, or accessible as specified.	3.11.1.1

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Number		Requirement Defectparagraph
Major		
188.	Enclosures of integral lubricant reservoirs not as specified.	3.11.1.2
189.	Fungus-and-moisture treatment not as specified.	3.11.2, 3.11.2.1
190.	Instruction plates not made and furnished as specified.	3.15
191.	Specified components not furnished by contractor.	3.16
192.	Casting and forgings not as specified.	3.17.1
193.	Metal fabrication not as specified.	3.17.2
194.	Bolted connections not as specified.	3.17.3
195.	Riveted connections not as specified.	3.17.4
196.	Welders not certified for welding as specified.	3.17.5.1
197.	Welded surface not prepared for welding as specified.	3.17.5.2
198.	Aluminum welding not as specified.	3.17.5.3
Minor		
201.	Finish not as specified.	3.13
202.	Color not as specified.	3.13
203.	Identification marking omitted or not as specified.	3.14

4.5.2 Tests.

4.5.2.1 Pumps. The pumps shall be tested for two hours as specified by NFPA 1901. Data shall be collected every 15 minutes. All gages shall be checked for accuracy of 10 percent of total gage range.

4.5.2.1.1 Hydrostatic. The entire pump, including the suction chamber and all piping, shall be tested to a hydrostatic pressure of 350 psi for 10 minutes. Nonconformance to 3.10.10.5 shall constitute failure of this test. This test shall be conducted after all tests specified in 4.5.2.4 have been completed.

4.5.2.2 Truck weight. The truck shall be weighted to determine curb weight and distribution of curb weight on front and rear axle. The imposed loading on front and rear axle will be computed using the curb weight, the operator weight at 175 pounds, and the payload uniformly distributed over the vehicle load area, to provide the specified gvw. Calculated imposed loads on front and rear axle will be utilized to ascertain that the suspension and tires furnished are of adequate capacity to meet specification requirements.

4.5.2.3 Environment tests.

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4.5.2.3.1 Test for ambient temperature and altitude conditions. When specified (see 6.2), the contractor shall perform a pumping test on the truck under actual conditions of the specified altitude above sea level and an ambient temperature of 125 °F. When pumping at heights above 2000 feet altitude is required, certification is required from the contractor that the vehicle will perform as specified in 3.10.1.5. The pumping test shall consist of the following:

a. Test the suction and priming.
b. The pumping test shall be for 3 hours, consisting of 2 hours at rated capacity at 150-psi discharge gage pressure, 1/2 hour at 70 percent of rated capacity at 200-psi discharge gage pressure, and 1/2 hour at 50 percent of rated capacity at 250-psi discharge gage pressure. The test shall be conducted with 20 feet of 3-inch suction hose and a suction lift of not less than 10 feet.

Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.4 Road tests. The loaded truck, including filled water tank, filled fuel tanks, lubricants, a 1500-pound, evenly distributed, simulated load, and a crew of three, weighing not less than 175 pounds each or equivalent deadweight, shall be tested as follows:

a. 100 miles over paved roads at maximum traffic speeds up to 55 mph.
b. 15 miles over graded dirt roads at speeds up to 30 mph.
c. Service brakes shall be applied during runs a. and b. above.
d. All lights, siren, horn, speedometer, hourmeter, controls, and instruments pertaining to the truck engine shall be tested.
e. The airbrake system shall be tested for conformance to 3.10.5.2.

Nonconformance shall constitute failure of these tests.

4.5.2.5 Operational tests. Each truck shall be loaded to its rated gvw, as specified in 3.4.2.1, and operated as follows:

a. 200 miles over paved roads at speeds up to 55 mph.
b. 50 miles over graded dirt roads at speeds up to 25 mph.
c. 20 miles off highway over cross-country terrain at speeds up to 15 mph. Cross-country terrain shall require intermittent operation of all-wheel drive. All-wheel drive or 4 x 4 mode shall be determined by operating the truck in the 4 x 2 mode until the truck looses traction; then the truck shall be shifted to the 4 x 4 mode and the operator shall be allowed 10 minutes to continue normal forward travel at not less than 10 mph. After 10 minutes, the truck shall be towed from the stuck condition and another attempt to travel through the same area shall be made. The same tire tracks shall not be used. The truck shall be put in the 4 x 4 mode and the attempt to traverse the area shall be made at a rolling start at any speed up to 15 mph. Failure to traverse the area on this attempt shall be cause for failure of this test.

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4.5.2.6 Electromagnetic interference. The truck shall be tested as specified in SAE J551. The manufacturer may furnish a certification in lieu of the test that the truck meets the requirements, together with a list of the suppression devices installed. Nonconformance to 3.10.19.1 and 3.12 shall constitute failure of this test.

4.5.2.7 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with 3.7.1 provisions shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing and marking of each truck and equipment shall be examined to determine compliance with the quality assurance provisions of MIL-STD-281 and MIL-E-10062, as applicable.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each complete truck with furnished equipment shall be preserved in accordance with MIL-STD-281, level A (mobile), with the following additions and exceptions.

5.1.1.1 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, type 1, method I, level A. The cooling system shall be preserved by the water and antifreeze procedures, except the antifreeze shall conform to MIL-A-46153.

5.1.1.2 Fire truck equipment. The fire truck equipment shall be preserved in accordance with the contractors recommendations that are adequate to prevent deterioration and damage during shipment and indeterminate storage, using the preservation criteria established in MIL-P-116. Specifically, the gasoline engine on the portable pump shall be preserved in accordance with the level A requirements of MIL-E-10062, type 1, method I.

5.1.1.3 Other equipment. Equipment shall be placed and secured as provided with the exception of exposed equipment subject to pilferage, damage or loss. This equipment shall be prepared as "on-vehicle equipment" in accordance with MIL-STD-281.

5.1.2 Level C. Each complete truck and all items issued with the truck shall be suitably preserved to afford protection against deterioration and damage from the contractor to the initial destination. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by use of preservative coatings, volatile corrosion inhibitors, and desiccated packs. Items requiring protection

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from physical damage to surfaces or which are fragile by nature shall be protected by wrapping pack, compartmentalization or cartonizing of the individual item. Exposed equipment subject to easy pilferage or loss shall be placed in secured compartments or truck cab, or in wooden boxes when practical. Wooden boxes, when used, shall be secured to the truck with metal strapping and in a manner to prevent damage to the painted surface of the truck. Items such as ladders that are too large to place in compartments, truck cab, or impractical to place in a wood box shall be secured in place using metal strapping. The cooling system shall be filled with a fresh, clean solution of 50-percent antifreeze conforming to MIL-A-46153 and a tag attached indicating the temperature to which the cooling system can be subjected before damage. The crankcase and all other lubrication reservoirs shall be filled to the operating level with the grade of military lubricant designed for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous location to indicate which military lubricant has been used and its grade and temperature range. A deprocessing guide shall be furnished indicating in detail the preservation material to be removed and other servicing required prior to placing the truck in operation.

5.2 Packing. Each complete truck, preserved as specified in 5.1, shall not require packing (see 6.6).

5.3 Marking. Marking (see 6.6) shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The truck is intended for initial attack in combating structural, forest, brush, grass, or similar natural cover fires.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. When a first article is required for inspection and approval, and numbers of trucks required (see 3.2).
d. When altitude modification kits are required (see 3.9.2 and 3.10.10).
e. When electromagnetic interference requirements shall be other than as specified (see 3.12).
f. When the color of the truck should be other than gloss lime-yellow (see 3.13.1).
g. When the welding will not be at the option of the contractor (see 3.17.5.2).
h. When altitude and high temperature pumping test shall be performed on the first article model (see 4.5.2.3.1).
i. Level of preservation required (see 5.1).

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6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one truck. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, tests, and approval of the first article test results and disposition of the first article.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 Ambient temperature. The temperature of the atmosphere measured at the intake side of the engine cooling radiator.

6.4.2 Excessive surging. Excessive surging shall be defined as any movement of the water or fuel that would cause the operator of the vehicle to lose control during acceleration, deceleration or any maneuver of the vehicle.

6.4.3 Flattening. Throughout the bend of a tube, the maximum diameter of the tube shall be equal to or greater than the tube's nominal diameter, and the tube's maximum diameter minus the tube's minimum diameter shall not be greater than 15 percent of the nominal tube diameter.

6.5 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of nonstandard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development, and Engineering Center (ATTN: STRBE-VF), lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under chapter 2 of MIL-HDBK-113.

6.6 Packing and marking. Levels of packing have not been included as they are not appropriate for this item. For purposes of preservation/packing-level marking, the marking shall be as required by the customer.

6.7 Subject term (key word) listing.

Firefighting truck
Fire truck
Truck, fire
Truck, fire fighting

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6.8 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in theses notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - ME

User activity:
Army - CE

Preparing activity:
Army - ME

Project 4210-A229

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-52984B(ME)	Truck, Fire Fighting: Brush, 1200 Gallon Water Tank, 6X6, Commercial Chassis

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FO[UT ALONG THIS LINE.)

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